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One Stone Two Birds—Enantioselective Bimetallic Catalysis for α-Amino Acid Derivatives with an Allene Unit

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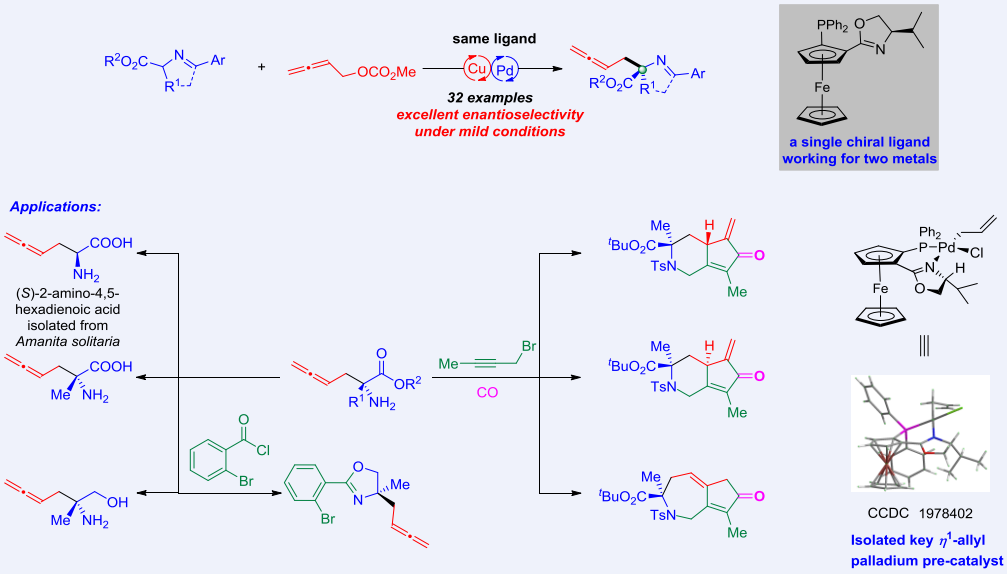
Keywords

Amino acids | Allene | Synergistic bimetallic Pd/Cu catalysis | Enantioselective synthesis | Mechanism

Main observation and conclusion

A highly enantioselective 2,3-allenylation of acyclic and cyclic α-imino carboxylates *via* a synergistic bimetallic Pd/Cu catalysis with the same commercially available (R,R_p)-^{i}Pr-FOXAP (also as Phosferrox, (*R*,*R*)-[2-(4′-*i*-propyloxazolin-2′-yl)ferrocenyl]diphenyl phosphine) ligand for both metals affording optically active 2,3-butadienyl α-amino acid derivatives in high to excellent yields with excellent enantioselectivities has been developed. The synthetic versatility of this reaction has been demonstrated by gram-scale synthesis, a catalytic enantioselective synthesis of naturally occurring (*S*)-2-amino-4,5-hexadienoic acid **A**, and conversions to several useful chemicals, such as optically active α-amino acids, β-amino alcohols, potential chiral oxazoline ligands bearing an allenic moiety, and bicyclic ketone compounds. A mechanism involving the roles of two metals and the single chiral ligand has been extensively studied based on the isolation of key palladium pre-catalyst and control experiments.

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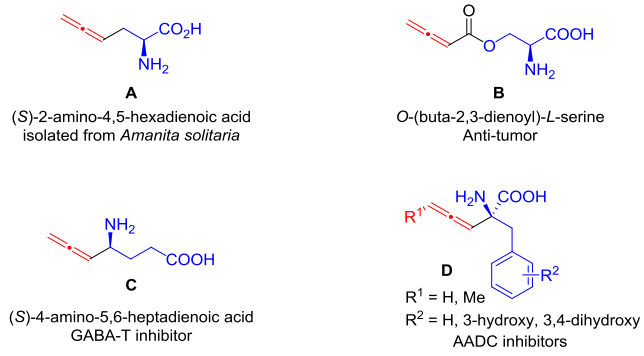
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Background and Originality Content

Amino acids show widespread applications in human nutrition, pharmaceutical arena, and the production of chiral catalysts.[1] Besides enzymatic methods in biosynthesis, enantioselective catalytic synthesis for diverse functionalized optically active non-natural amino acids is an alternative pathway.[2] On the other hand, during the last two decades, allenes have become more and more irreplaceable in organic chemistry and medicinal chemistry due to their unique physical,[3] biological,[4] and chemical properties.[5] Not only widely present in naturally occurring molecules,[4] allenic structures have been also demonstrated to be very useful building blocks in synthetic transformation and construction of complex molecules.[5] In fact, molecules possessing both allene and amino acid units are extremely useful and attractive due to their highly functionalized properties for synthetic organic chemists and biology-related scientists: allene-substituted amino acids are widely distributed in many fungi and considered as a common intermediate in the biosynthesis of various norleucine-type amino acids.[6] For example, (*S*)-2-amino-4,5-hexadienoic acid (Scheme 1, **A**) was first reported and isolated from the fruiting bodies of *Amanita solitaria* in 1968.[6a] In addition, allenic amino acids indeed exhibit potent bioactivity: *O*-(buta-2,3-dienoyl)-*L*-serine (Scheme 1, **B**) shows anti-tumor activity against murine L1210 leukemia;[7] there are also some allenic amino acids proved to be enzyme inhibitors, such as a GABA-T inhibitor[8] (Scheme 1, **C**) and aromatic amino acid decarboxylase inhibitors (Scheme 1, **D**).[8-9] Thus, development of powerful methods for highly enantioselective synthesis of optically active molecules containing both an amino acid unit and an allene moiety is of great interest.

Scheme 1 The importance of allene-substituted amino acids

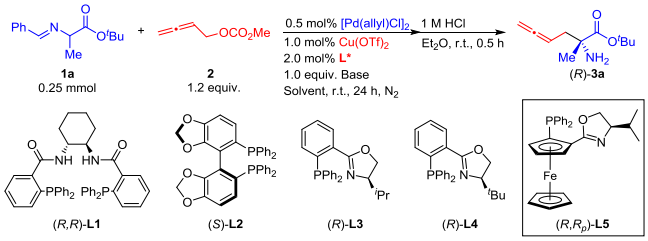


We reasoned that the reaction of methylene-π-allyl palladium[10] with α-imino carboxylates[11] in the presence of a base would be a straightforward strategy to afford α-imino-α-allenyl-carboxylates, which are precursors for α-amino acids bearing an allene moiety. When we were developing such a reaction, Wang *et al*. reported such a reaction with two different ligands for two metals.[12] After facing the Covid-19 and addressing some data discrepancy, herein, we wish to report a highly active synergistic Pd/Cu dual catalysis[13-15] for enantioselective 2,3-butadienylation of acyclic and cyclic α-imino carboxylates with a single chiral ligand, *i.e*., commercially available (*R*,R_p)-^{i}Pr-FOXAP, for both metals with a very wide scope. The roles of each metal and the chiral ligand have been carefully unveiled based on the isolation of key palladium pre-catalyst.

Results and Discussion

We began our study with acyclic α-imino carboxylate **1a** and buta-2,3-dienyl methyl carbonate **2** as model substrates. With

Table 1 The effect of ligands, bases, and solvents[a]



Entry	L*	Base	Solvent	1a NMR Recovery[b]/%	(*R*)-3a Yield/%	(*R*)-3a *ee*/%
1[c]	(*R*,*R*)-**L1**	Cs_2CO_3	THF	74	18	27
2	(*S*)-**L2**	Cs_2CO_3	THF	6	70	22
3	(*R*)-**L3**	Cs_2CO_3	THF	48	32	-36
4	(*R*)-**L4**	Cs_2CO_3	THF	50	32	-49
5	**(*R*,R_p)-L5**	**Cs_2CO_3**	**THF**	**-**	**87**	**93**
6	(*R*,R_p)-**L5**	Li_2CO_3	THF	98	-	-
7	(*R*,R_p)-**L5**	Na_2CO_3	THF	> 99	-	-
8	(*R*,R_p)-**L5**	K_2CO_3	THF	69	25	95
9	(*R*,R_p)-**L5**	Rb_2CO_3	THF	42	46	89
10	(*R*,R_p)-**L5**	Cs_2CO_3	Et_2O	99	-	-
11	(*R*,R_p)-**L5**	Cs_2CO_3	MTBE	78	18	72
12	**(*R*,R_p)-L5**	**Cs_2CO_3**	**DCM**	**-**	**86**	**97**
13	(*R*,R_p)-**L5**	Cs_2CO_3	Toluene	87	-	-

[a] Reaction conditions: **1** (0.25 mmol), **2** (0.3 mmol), Pd/**L***(1.0 mol%), Cu/**L*** (1.0 mol%), base (0.25 mmol), and solvent (0.5 mL) at rt under N_2 atmosphere for 24 h. Isolated yield. The *ee* were determined by chiral HPLC analysis after column chromatographic separation on silica gel. [b] Determined by ^{1}H NMR analysis before workup with 1 M HCl using CH_3NO_2 as the internal standard. [c] 1.0 mol% of (*R*,*R*)-**L1** was used.

Cs_2CO_3 as the base and tetrahydrofuran (THF) as the solvent, a [Pd/**L***+Cu/**L***] catalyst combination was applied to identify the optimal single chiral ligand for both metals. The most commonly used ligands for enantioselective allenylation reactions failed to give satisfactory results: the use of Trost ligand (*R*,*R*)-**L1** gave (*R*)-**3a** in only 18% yield with 27% ee (Table 1, entry 1), and biphenyl ligand (*S*)-**L2** resulted in a higher yield, but still with a low ee (Table 1, entry 2). It was gratifying to observe that the enantiocontrol was getting better by using phosphine oxazoline (PHOX) ligands: (*R*)-**L3** resulted in 32% yield with 36% ee (Table 1, entry 3). By tuning the steric effect, it was observed that (*R*)-**L4** improved the ee to 49% (Table 1, entry 4). To our delight, phosferrox (*R*,R_p)-**L5** was able to provide 87% yield and 93% ee of (*R*)-**3a** with no recovery of **1a** as monitored by ^{1}H NMR analysis (Table 1, entry 5). Next, the effect of base was investigated. A series of inorganic carbonates were screened to show that the reactivity was improved by increasing the radius of the cation (Table 1, entries 5—9). The solvent effect was also studied: ethyl ether failed to give (*R*)-**3a** with nearly quantitative recovery of **1a** (Table 1, entry 10); in methyl *tert*-butyl ether (MTBE), the reaction exhibited a very low reactivity and enantioselectivity (Table 1, entry 11). Excitingly, the reaction in dichloromethane (DCM) afforded (*R*)-**3a** in 97% ee with a yield of 86% (Table 1, entry 12). Toluene was not effective for the transformation (Table 1, entry 13).

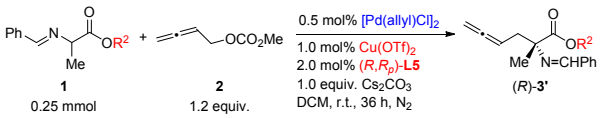
With the optimized conditions in hand, we next examined the scope of this straightforward approach for enantioselective 2,3-butadienylation of α-imino carboxylates. To our delight, substrates with commonly used R^2 groups (^{t}Bu, Me, Et, and ^{i}Pr, Table 2, entries 1—4) reacted smoothly, indicating that the steric hindrance in R^2 group did not have remarkable influence on this enantioselective catalytic process.

Furthermore, we turned our attention to investigate the effects of R^1 in α-imino carboxylates (Table 3). The α-imino carboxylates with a linear alkyl and branched alkyl substitution all reacted smoothly with buta-2,3-dienyl methyl carbonate **2** to afford optically active amino derivatives with an allene unit **3** in high to excellent yields with excellent ee, indicating that the protocol

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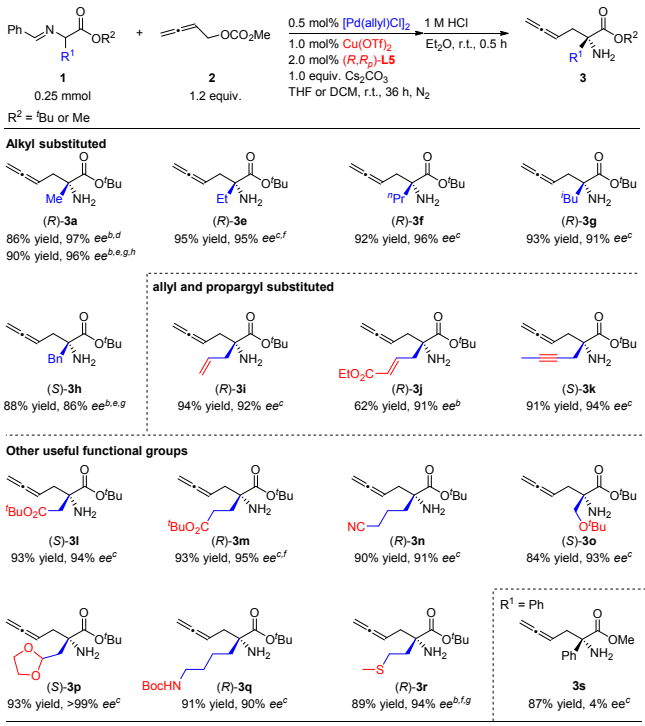
Table 2 The effect of R^2 groups in α-imino carboxylates **1**[a]



Entry	R^2	(*R*)-**3'**	
		Yield/%	*ee*/%
1	^{t}Bu (**1a**)	92	94((*R*)-**3'a**)
2	Me (**1b**)	91	94((*R*)-**3'b**)
3	Et (**1c**)	92	96((*R*)-**3'c**)
4	^{i}Pr (**1d**)	87	94((*R*)-**3'd**)

[a] Reaction conditions: **1** (0.25 mmol), **2** (0.3 mmol), Pd/(*R*,R_p)-**L5** (1.0 mol%), Cu/(*R*,R_p)-**L5** (1.0 mol%), Cs_2CO_3 (0.25 mmol), and DCM (0.5 mL) at rt under N_2 atmosphere for 36 h. Isolated yield. The *ee* was determined by chiral HPLC analysis after column chromatographic separation on silica gel.

Table 3 Substrate scope of acyclic α-imino carboxylates **1**[a]

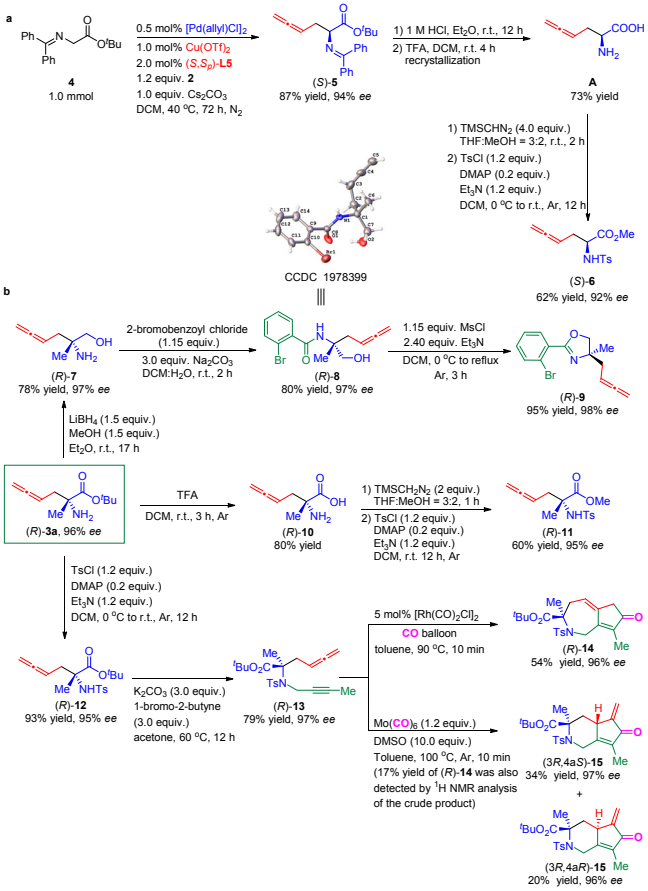


[a] Reaction conditions: **1** (0.25 mmol), **2** (0.3 mmol), Pd/(*R*,R_p)-**L5** (1.0 mol%), Cu/(*R*,R_p)-**L5** (1.0 mol%), Cs_2CO_3 (0.25 mmol), and solvent (0.5 mL) at rt under N_2 atmosphere for 36 h. Isolated yield. The *ee* was determined by chiral HPLC analysis after column chromatographic separation on silica gel. [b] DCM as solvent. [c] THF as solvent. [d] 24 h. [e] 48 h. [f] 60 h. [g] 40 °C. [h] The reaction was conducted with 8.0 mmol of **1a**, and 9.6 mmol of **2** to afford 1.4206 g (*R*)-**3a**.

was amenable to different carbon chain commonly existing in bioactive amino acid derivatives (Table 3, (*R*)-**3a** to (*R*)-**3g**). When R^1 = Bn, (*S*)-**3h** was generated in 88% yield with 86% ee at 40 °C. In addition, **1i** and **1k** bearing allyl and but-2-ynyl groups worked well in this transformation affording (*R*)-**3i** and (*S*)-**3k** in ee of 92% and 94%, respectively. Substrate **1j** with an electron-deficient C=C bond also gave (*R*)-**3j** in 62% yield and 91% ee. Last but not the least, a series of synthetically attractive functional groups, such as ester ((*S*)-**3l** and (*R*)-**3m**), cyano ((*R*)-**3n**), ether ((*S*)-**3o**), acetal ((*S*)-**3p**), *tert*-butoxycarbonyl amino ((*R*)-**3q**), and methyl thioether ((*R*)-**3r**), were well tolerated, proving the protocol a viable tool for further syntheses of complicate bioactive compounds bearing an allenic moiety. However, the reaction of phenylglycine methyl ester derived α-imino carboxylate resulted in the formation of **3s** in 87% yield with 4% ee. The practicality was also demonstrated by applying 8.0 mmol of **1a** and 9.6 mmol of **2**, affording 1.4206 g of (*R*)-**3a** in 90% yield and 96% ee at 40 °C for 48 h.

In addition, ketimine Schiff base **4** with R^1 = H also reacted smoothly with buta-2,3-dienyl methyl carbonate **2** under the standard condition to afford (*S*)-**5** in 87% yield with 94% ee (Scheme 2a). After removing both protecting groups with an aqueous hydrogen chloride solution (1 M HCl) and trifluoroacetic acid (TFA), subsequently, the naturally occurring (*S*)-2-amino-4,5-hexadienoic acid (**A**) listed in Scheme 1 was formed and isolated easily *via* simple recrystallization in 73% yield. The ee value was carefully determined by HPLC to be 92% by its conversion to the corresponding methyl ester (*S*)-**6**.

Scheme 2 (a) Catalytic enantioselective synthesis of naturally occurring (*S*)-2-amino-4,5-hexadienoic acid. (b) Synthetic applications of (*R*)-**3a** and establishment of stereochemistry



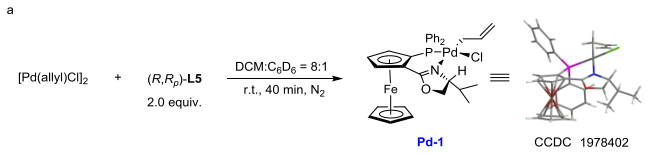
Furthermore, we performed the reduction of (*R*)-**3a** to afford amino alcohol (*R*)-**7**, which was further reacted with 2-bromobenzoyl chloride, yielding (*R*)-**8** with an aryl bromide unit in 80% yield with 97% ee. Luckily, the absolute configuration of (*R*)-**8** was eventually established by the X-ray crystallography, which also unambiguously established the stereochemistry of this reaction (Scheme 2b). Oxazolination of (*R*)-**8** was successfully realized in the presence of methanesulfonyl chloride (MsCl) to afford the oxazolinyl allene product (*R*)-**9** (95% yield and 98% ee), which may be used as a chiral ligand directly or further transformed to chiral phosphine oxazoline (PHOX) ligands bearing allenic structure as a coordinative regulator due to its unique coordination behaviour.[16] Furthermore, we performed hydrolysis and tosylation of (*R*)-**3a** to afford amino acid (*R*)-**10** and tosylamide (*R*)-**12**, respectively, in high yields with excellent ee. In addition, (*R*)-**12** easily reacted with 1-bromo-2-butyne to afford alkyne-allene (*R*)-**13** in 79% yield with 97% ee. With $[Rh(CO)_2Cl]_2$ as catalyst, [5.3.0]bicyclopentenone compound (*R*)-**14** was formed by the Pauson-Khand reaction of (*R*)-**13** in 54% yield with 96% ee. (*R*)-**13** could also be converted to [4.3.0]bicyclic products **15** with 1.2 equiv. of $Mo(CO)_6$ as the

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source of carbonyl group. These bicyclic structures are important common skeletons in natural products.[17]

To unveil the nature of this Pd/Cu dual catalytic process, the reaction of $[Pd(allyl)Cl]_2$ and (R,R_p)-**L5** was conducted in the mixture of DCM/C_6D_6 (V/V = 8 : 1) at rt for 40 min and (R,R_p)-^{i}Pr-FOXAP-η^1-allyl palladium chloride **Pd-1** has been isolated with its structure determined by X-ray diffraction study (Figure 1a). Control experiments were then conducted to probe the reaction mechanism: The reaction performed with 1 mol% of $Cu(OTf)_2$ and 1 mol% of (R,R_p)-**L5** failed to afford (*R*)-**3a**, indicating the critical role of palladium for this 2,3-butadienylation process (Figure 1b, entry 1). When the reaction was conducted in the absence of $Cu(OTf)_2$ with only 0.5 mol% of $[Pd(allyl)Cl]_2$ and 1 mol% of (R,R_p)-**L5**, the conversion of **1a** was only 16% and the yield for (*R*)-**3a** was 9% with an enantioselectivity as low as 14% (Figure 1b, entry 2). When the reaction was performed with 0.5 mol% of $[Pd(allyl)Cl]_2$, 1 mol% of $Cu(OTf)_2$, and 2 mol% of (R,R_p)-**L5**, (*R*)-**3a** was yielded in 86% yield with 97% ee (Figure 1b, entry 3). Meanwhile, the combination of 1 mol% of **Pd-1**, 1 mol% $Cu(OTf)_2$, and 1 mol% of (R,R_p)-**L5** reacted smoothly to afford (*R*)-**3a** in 87% yield with 97% ee (Figure 1b, entry 4). These data indicate that **Pd-1** served as the pre-catalyst in the catalytic cycle. When the reaction was performed with 1 mol% of **Pd-1** and 1 mol% of $Cu(OTf)_2$ in the absence of extra chiral ligand, (*R*)-**3a** was yielded in only 13% yield with 68% ee and 85% of **1a** was recovered as determined by ^{1}H NMR analysis. These data indicate that coordination of (R,R_p)-**L5** to $Cu(OTf)_2$ must have resulted in the formation of a highly reactive chiral nucleophile for the improved reactivity and enantioselective control (Figure 1b, compare entry 4 with entry 5).



Entry	Pd (*x* mol%)	$Cu(OTf)_2$ (*y* mol%)	(R,R_p)-**L5** (*z* mol%)	Time/h	Conv. of **1a**/%	Yield of (*R*)-**3a**/%	Ee of (*R*)-**3a**/%
1	-	1	1	24	< 1	-	-
2	$[Pd(allyl)Cl]_2$ (0.5)	0	1	24	16	9	14
3	$[Pd(allyl)Cl]_2$ (0.5)	1	2	24	>99	86	97
4	**Pd-1** (1)	1	1	36	>99	87	97
5	**Pd-1** (1)	1	0	36	15	13	68

Figure 1 The studies on reaction mechanism. (a) Preparation of (R,R_p)-^{i}Pr-FOXAP-η^1-allylpalladium chloride **Pd-1**. (b) Control experiments.

Based on these studies, a bimetallic catalytic cycle is proposed as shown in Figure 2. In the presence of acyclic α-imino carboxylates **1a** and base, the catalytically active species L*Pd(0) would be generated *in situ* from pre-catalyst **Pd-1**. This chiral palladium complex L*Pd(0) would react with buta-2,3-dienyl methyl carbonate **2** *via* S_N2'-type oxidative addition to generate η^1-1,3-dienyl Pd(II) intermediate **II**. Subsequent σ-π rearrangement would yield the delocalized methylene-π-allyl palladium(II) intermediate **III**. With the chiral copper(II) complex and base, α-imino carboxylate **1a** would be readily converted to *N*-metalated azomethine ylide **IV**.[11i] The highly reactive nucleophile **IV** would attack the methylene-π-allyl palladium(II) **III** in a highly enantioselective manner to afford the 2-(*N*-imino)-4,5-allenoates and regenerate both chiral Pd and Cu catalysts.

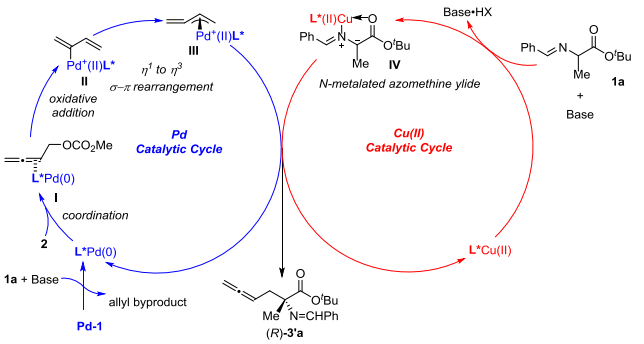
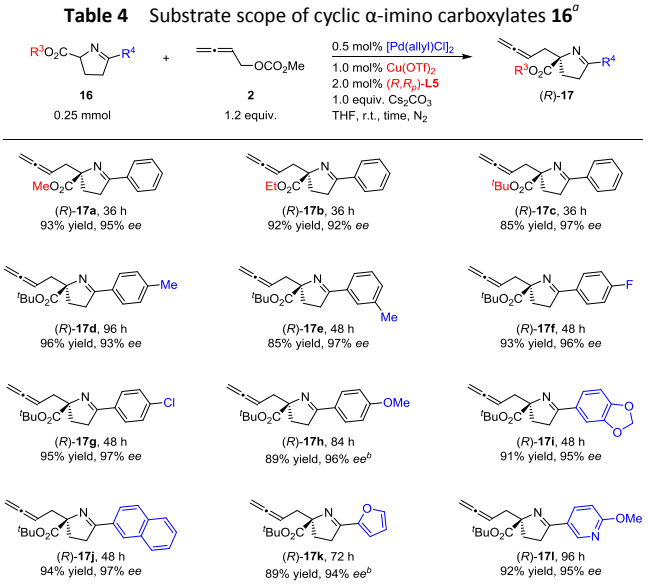


Figure 2 Proposed bimetallic catalytic cycles.

Cyclic α-imino carboxylates are privileged building blocks[18] in organic chemistry and have been widely applied in the construction of bioactive nitrogen-containing heterocyclic molecules.[19] Furthermore, such products could be reduced to non-natural proline derivatives with α-quaternary stereogenic center. It is interesting to see that the current mono-chiral ligand based Pd/Cu dual catalytic process is also amenable to the enantioselective 2,3-butadienylation of cyclic α-imino carboxylates. Different R^3 in the ester groups (Table 4, (*R*)-**17a** to (*R*)-**17c**) and aryl or heteroaryl substitutions (R^4) (Table 4, (*R*)-**17d** to (*R*)-**17l**) all worked well, resulting in excellent yields with ee (85%—96% yields, 92%—97% ee), providing a reliable strategy for the synthesis of extra optically active non-natural proline derivatives containing allenic structures. The absolute configurations of the chiral centers in **17** are tentatively assigned to be *R* based on the analogy to (*R*)-**3a**.

Table 4 Substrate scope of cyclic α-imino carboxylates **16**[a]



[a] Reaction conditions: **16** (0.25 mmol), **2** (0.3 mmol), Pd/(R,R_p)-**L5** (1.0 mol%), Cu/(R,R_p)-**L5** (1.0 mol%), Cs_2CO_3 (0.25 mmol), and THF (0.5 mL) at rt under N_2 atmosphere. Isolated yield. The *ee* were determined by chiral HPLC analysis after column chromatographic separation on silica gel. [b] 40 °C.

Conclusions

In summary, we have developed a highly enantioselective 2,3-butadienylation of acyclic and cyclic α-imino carboxylates *via* a synergistic Pd/Cu dual catalysis with (R,R_p)-^{i}Pr-FOXAP as the ligand for both metals. A series of chiral 2,3-butadienyl α-amino acid derivatives bearing versatile functionalities were easily synthesized in high to excellent yields with enantioselectivities. This reaction was easily scaled up and successfully applied in the catalytic enantioselective synthesis of naturally occurring (*S*)-2-amino-

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4,5-hexadienoic acid **A**. In addition, the prepared 2-amino-4,5-allenoates could be easily transformed to synthetically versatile compounds such as optically active 2,3-butadienyl α-amino acids, β-amino alcohols, oxazolines bearing an allenic moiety, and bicyclic ketone compounds. Further applications of this bimetallic catalysis strategy in stereocontrol on both axial chirality of allene and central chirality of α-imino carboxylates, applications in ligand-based asymmetric catalysis/natural products, and study on biological activity are actively pursued in both laboratories of Ma and Zhang.

Experimental

Preparation of chiral Pd catalyst

To an oven-dried 4 mL vial were added $[Pd(allyl)Cl]_2$ (1.8 mg, 0.005 mmol), (R,R_p)-^{i}Pr-FOXAP (4.8 mg, 0.01 mmol), and freshly distilled solvent (1 mL) in a glove box. The resulting mixture was stirred at room temperature for 40 min.

Preparation of chiral Cu catalyst

To an oven-dried 4 mL vial were added $Cu(OTf)_2$ (3.6 mg, 0.01 mmol), (R,R_p)-^{i}Pr-FOXAP (4.8 mg, 0.01 mmol), and freshly distilled solvent (1 mL) in a glove box. The resulting mixture was stirred at room temperature for 40 min.

Typical procedure for the synthesis of (*R*)-3a

To an oven-dried Schlenk flask were added Cs_2CO_3 (81.5 mg, 0.25 mmol), **1a** (58.6 mg, 0.25 mmol), chiral Pd catalyst (0.25 mL), chiral Cu catalyst (0.25 mL), and **2** (38.6 mg, 0.3 mmol) sequentially in a glove box. After being stirred vigorously at room temperature for 24 h, the reaction was complete as monitored by TLC. 2.0 mL of Et_2O was added and the resulting mixture was then filtered through a short pad of silica gel eluted with Et_2O (50 mL). After removal of the solvent under vacuum, 2 mL of Et_2O and 2 mL of 1 M HCl were added sequentially. After the resulting mixture was stirred at room temperature for 0.5 h, 4 mL of Et_2O was added. The pH value of the mixture was regulated to 8—9 with solid K_2CO_3 (*ca*. 450 mg). The organic layer was separated and the aqueous layer was extracted with Et_2O (4 mL × 4). The combined organic layer was dried over anhydrous Na_2SO_4, decanting into a round-bottom flask, and concentrated. After removal of the solvent under vacuum, the residue was purified by flash chromatography on silica gel to afford (*R*)-**3a** (42.8 mg, 86% yield, 97% ee) [eluent: petroleum ether: acetone = 20 : 1 (168 mL), petroleum ether: acetone = 2 : 1 (225 mL)]: oil; HPLC conditions: AD-H column, hexane/^{i}PrOH = 100/1, 1.0 mL/min, λ = 214 nm, t_R (minor) = 10.9 min, t_R (major) = 11.9 min; $[\alpha]_D^{28}$ + 28.2 (*c* 0.54, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 5.03 (quintet, J = 7.3 Hz, 1H, =CH), 4.68 (dt, J_1 = 6.8 Hz, J_2 = 2.8 Hz, 2H, $=CH_2$), 2.50—2.40 (m, 1H, one proton of CH_2), 2.25—2.16 (m, 1H, one proton of CH_2), 1.81—1.67 (brs, 2H, NH_2), 1.46 (s, 9H, $OC(CH_3)_3$), 1.30 (s, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 209.9, 176.2, 84.9, 80.9, 74.5, 58.0, 40.1, 27.9, 26.3; IR (neat) ν: 3379, 2977, 1956, 1722, 1368, 1157, 1112 cm^{-1}; MS (ESI) m/z (%): 198 ($M+H^+$); HRMS Calcd. for $C_{11}H_{20}NO_2^+$ ($M+H^+$): 198.1489, found 198.1487.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202100002.

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